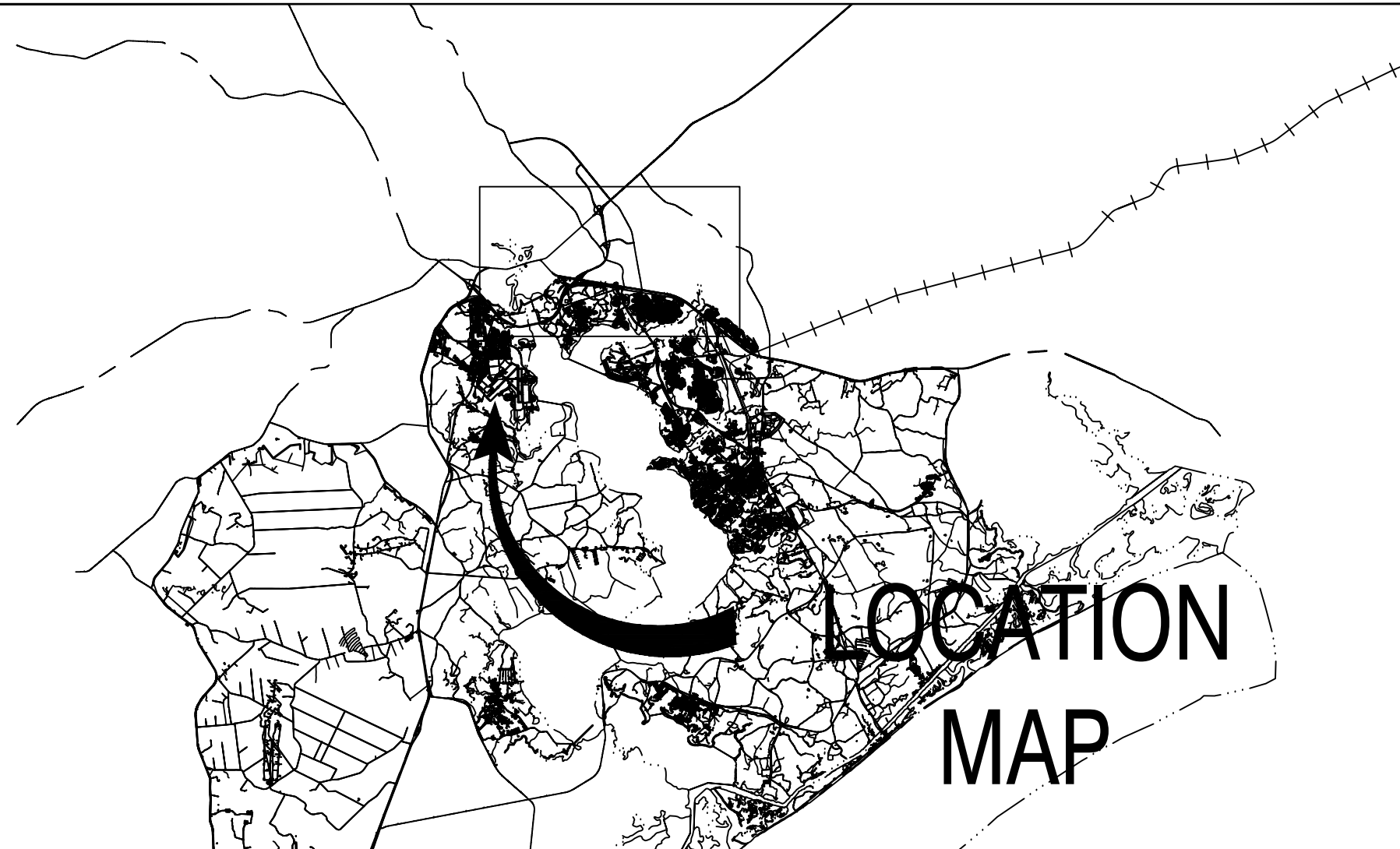


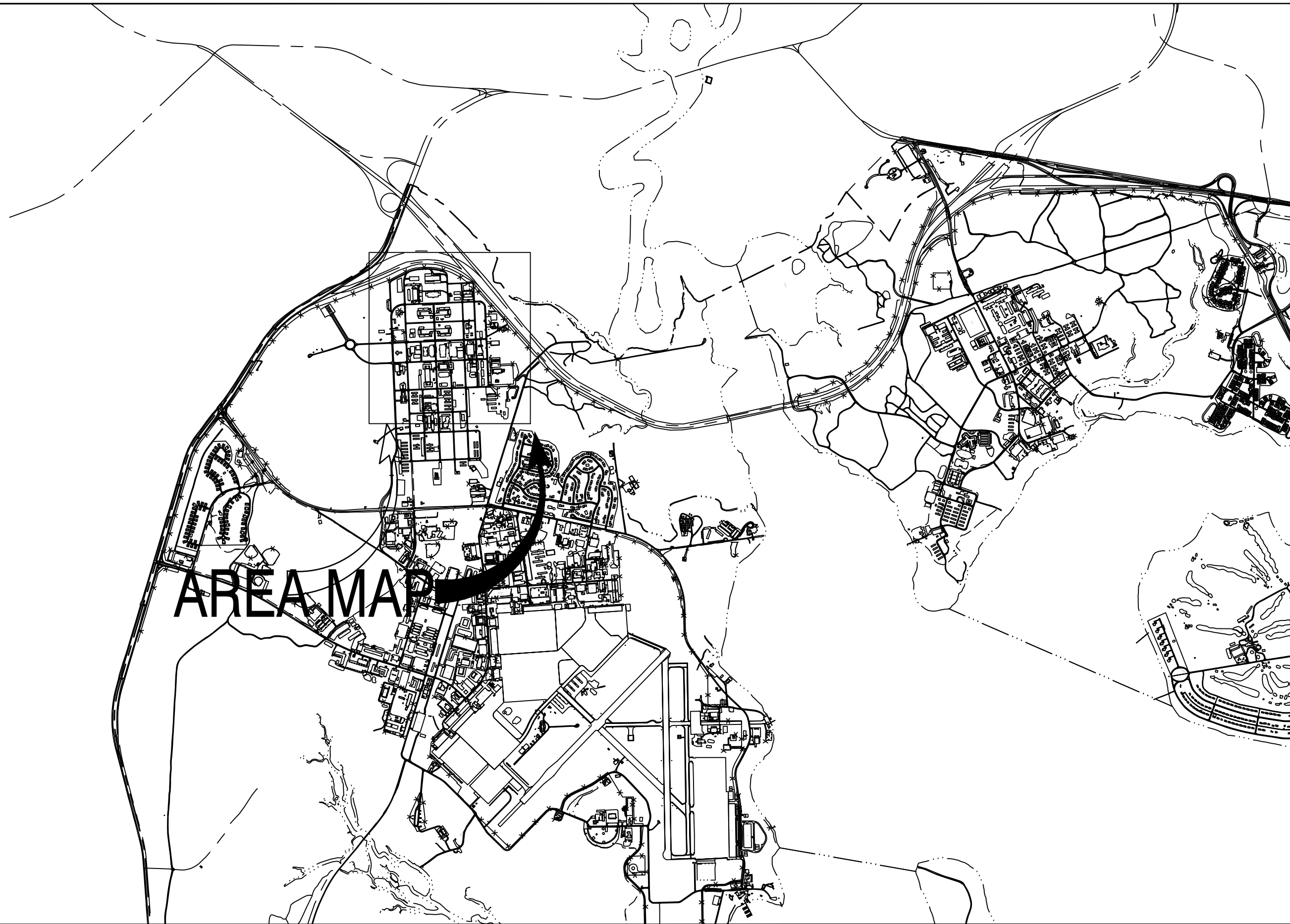
VICINITY MAP



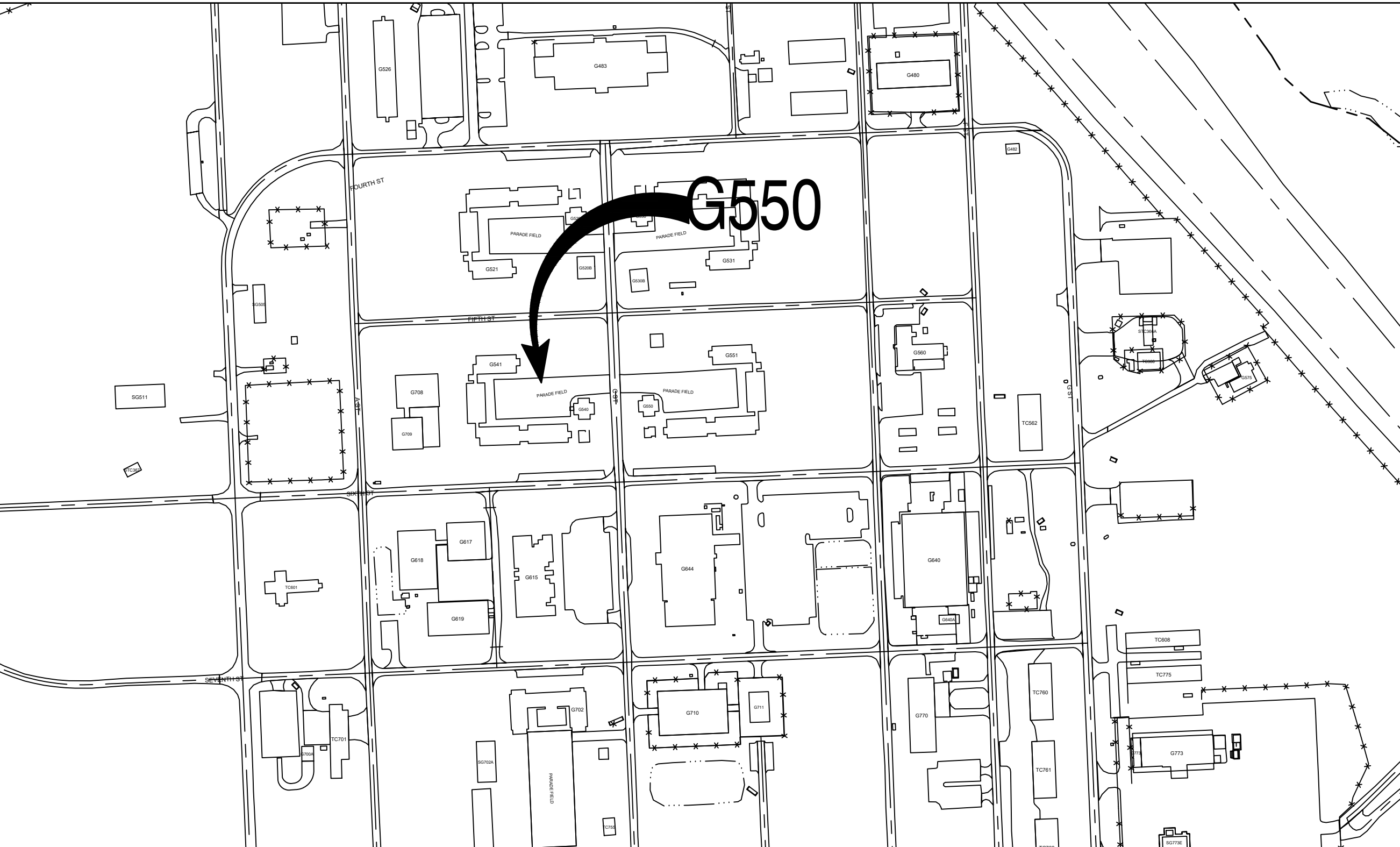
25-0080 GEIGER QUAD G550 DOAS REPLACEMENT

MARINE CORPS BASE, CAMP LEJEUNE, N.C.

LOCATION MAP



AREA MAP



GENERAL NOTES

1. THESE DRAWINGS INDICATES GENERAL CONDITIONS WITH INFORMATION COMPILED FROM FIELD OBSERVATION. CONSTRUCTION AND MATERIALS INDICATED ARE GENERAL IN NATURE AND ARE NOT INTENDED TO FULLY DESCRIBE THE EXISTING CONSTRUCTION PRESENT. THE CONTRACTOR IS RESPONSIBLE TO INSPECT ALL FACILITIES TO BE DEMOLISHED TO HIS OWN SATISFACTION AND INCLUDE ALL NECESSARY COSTS TO COMPLETELY REMOVE EACH FACILITY AS DESCRIBED. CONTRACTOR SHALL BE RESPONSIBLE FOR ACTUAL FIELD VERIFICATION PRIOR TO BIDDING, ORDERING MATERIALS AND DURING EVERY STEP OF CONSTRUCTION FOR EXISTING SURFACES, DIMENSIONS AND CONDITIONS. NUMBER OF ITEMS AND LOCATIONS SHOWN ARE APPROXIMATE. CONTRACTOR MUST VERIFY ALL CONDITIONS AND INCLUDE IN SCOPE OF WORK ACCORDINGLY.
2. PRIOR TO STARTING DEMOLITION ON ANY STRUCTURES OR UTILITIES, THE CONTRACTOR SHALL FAMILIARIZE HIMSELF WITH THE EXISTING CONDITIONS OF ANY STRUCTURES AND UTILITIES, THE CONTRACTOR SHALL DEVELOP A PLAN OF DEMOLITION THAT ENSURES ALL ACTIVITIES ARE COMPLETED IN A SAFE MANNER. PROVIDE ANY TEMPORARY SHORING, SHEETING OR SUPPORT REQUIRED TO COMPLETE WORK IN A SAFE MANNER.
3. COMPLETELY REMOVE ALL STRUCTURE AND UTILITIES INDICATED, BOTH ABOVE GROUND AND BELOW GROUND.
4. WHERE ROADS, SIDEWALKS, ETC ARE INDICATED TO BE CUT AND PATCHED, EACH SHALL BE REMOVED AND REPLACED ALONG NEAT SAWCUT LINES, AND TO THE NEAREST JOINT WHERE IT EXISTS.
5. THIS AREA OF WORK IS LOCATED WITHIN RESTRICTED AREAS. ACCESS TO THIS AREA SHALL BE COORDINATED WITH THE CONTRACTING OFFICER.
6. REPAIR AND REFINISH AREAS DAMAGED OR DISTURBED BY NEW WORK TO BE PROVIDED UNDER THIS CONTRACT.
7. DIMENSIONS PROVIDED ON EXISTING ITEMS ARE PROVIDED TO AID THE CONTRACTOR UNDERSTAND THE SCOPE OF WORK. FIELD VERIFY ALL CONDITIONS. ALL THE DIMENSIONS SHOWN ARE APPROXIMATE ONLY.
8. METHOD OF REMOVAL OF ANY BUILDING COMPONENT SHALL BE APPROVED BY THE CO CONTRACTING REPRESENTATIVE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ENSURING THAT REMOVAL OF EXISTING COMPONENTS DO NOT AFFECT ADJACENT OCCUPIED AREAS.
9. DEMOLITION OPERATIONS SHALL BE COMPLETED IN COMPLIANCE WITH ALL STATE AND FEDERAL REGULATIONS AND AS SPECIFIED.
10. ALL CONSTRUCTION ACTIVITIES SHALL BE IN ACCORDANCE WITH OSHA STANDARDS, AND GOVERNMENT REQUIREMENTS
11. THE NOTATION (TYP) INDICATES THAT THE CONDITION OCCURS IN MORE THAN ONE PLACE INCLUDING ALL SIMILAR CONDITIONS NOT INDIVIDUALLY NOTED.
12. COORDINATE STAGING AREA, SITE ACCESS, TOILET USE WITH THE CO CONTRACTING REPRESENTATIVE.
13. CONTRACTOR SHALL BE RESPONSIBLE FOR SITE SECURITY AND JOB SAFETY.
14. AREAS OUTSIDE THE LIMITS OF PROPOSED WORK DISTURBED BY THE CONTRACTOR'S OPERATIONS SHALL BE RESTORED BY THE CONTRACTOR TO THEIR ORIGINAL CONDITION AT THE CONTRACTOR'S EXPENSE AS SOON AS PRACTICABLE.
15. CONTRACTOR SHALL MINIMIZE DUST, SEDIMENT AND DEBRIS FROM EXITING THE CONSTRUCTION SITE AND SHALL BE RESPONSIBLE FOR CLEANUP, REPAIRS AND CORRECTIVE ACTION IF SUCH OCCURS. CONTRACTOR SHALL DISPOSE OF DEBRIS IN ACCORDANCE WITH APPLICABLE FEDERAL, STATE, AND LOCAL REGULATIONS, ORDINANCE AND STATUTES.
16. BURNING WILL NOT BE ALLOWED ON THE PROJECT SITE UNLESS AUTHORIZED IN WRITING BY THE CONTRACTING OFFICER.
17. ADJACENT STRUCTURES AND UTILITIES MUST REMAIN IN OPERATION DURING DEMOLITION ACTIVITIES. EXISTING ROADS SHALL REMAIN OPEN AND ACCESSIBLE BY VEHICULAR AND PEDESTRIAN TRAFFIC. IF ROADWAY CLOSURE IS REQUIRED, APPROVAL SHALL BE SECURED FROM THE CONTRACTING OFFICER. THE CONTRACTOR SHALL PROVIDE BARRICADES, LIGHTS, SIGNAGE AND OTHER PROTECTIVE DEVICES IN ACCORDANCE
18. CONTRACTOR'S PERSONNEL MUST WEAR IDENTIFICATION AT ALL TIMES.
19. COORDINATE POWER OUTAGES WITH THE GOVERNMENT. MAINTAIN EXISTING ELECTRICAL SYSTEMS IN SERVICE. DISABLE SYSTEMS ONLY TO MAKE SWITCHOVERS AND CONNECTIONS. OBTAIN PERMISSION FROM THE CONTRACTING OFFICER AT LEAST 5 DAYS BEFORE PARTIALLY OR COMPLETELY DISABLING SYSTEM. MINIMIZE OUTAGE DURATION.

DRAWING INDEX

NAVFAC #	SHEET #	SHEET NAME
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60042370	M101	MECHANICAL SITE PLAN
60042371	M501	MECHANICAL DETAILS AND SCHEDULE
60042372	M801	MECHANICAL BARRACKS DOAS DDC DIAGRAM
60042373	M802	MECHANICAL BARRACKS DOAS DDC POINTS
60042374	M803	MECHANICAL ADMIN DOAS DDC DIAGRAM
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REVISIONS

SYM	REVISIONS	DATE	APPROVED
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△			
△			

CONSTRUCTION DOCUMENTS

T001

DES. C. HUSKEY		MARINE CORPS BASE	
DR. C. HUSKEY		CAMP LEJEUNE, NORTH CAROLINA	
CHK. C. MAAS		GEIGER QUAD G550	
SUBMITTED BY: C. MAAS		DOAS REPLACEMENT	
DESIGN DIR. C. MAAS		TITLE SHEET ABBREVIATIONS, AREA MAPS LEGEND	
APPROVED: PWO OR OICC:	DATE:	SIZE: E1	CODE IDENT. NO: 80091
SATISFACTORY TO:		NAVFAC DRAWING NO. 60042368	
24 FEBRUARY 2026		CONST. CONTR. NO:	
SCALE: NOTED		SPEC NO.: 25-0080	
		SHEET: 1 OF 8	

MECHANICAL GENERAL NOTES:

- ALL MECHANICAL WORK MUST BE IN STRICT COMPLIANCE WITH ALL APPLICABLE FEDERAL, STATE AND LOCAL CODES AND STANDARDS.
- ALL DIMENSIONS AND ELEVATIONS FOR NEW EQUIPMENT, DUCTWORK, PIPING AND APPARATUS ARE APPROXIMATE AND ARE ONLY FOR CONTRACTOR'S GUIDANCE. CONTRACTOR MUST SUBMIT DIMENSIONS AND ELEVATIONS VERIFIED IN THE FIELD. DUCTWORK AND PIPING INDICATED ON THE DRAWINGS, SECTIONS AND PROSPECTIVE VIEWS ARE SHOWN DIAGRAMMATICALLY. DUCT AND PIPE ELEVATIONS IN EXACT LOCATIONS MUST BE DETERMINED BY THE INSTALLING CONTRACTOR AND DETAILED ON THE SHOP DRAWINGS.
- ALL DUCT DIMENSIONS INDICATED ON PLAN ARE CLEAR INSIDE DIMENSIONS. CONTRACTOR MUST ACCOUNT FOR THE THICKNESS OF EXTERIOR INSULATION WHEN DETERMINING INSTALLATION CLEARANCES.
- THE CONTRACTOR MUST TEMPORARILY COVER ALL EXPOSED DUCT AND PIPE OPENINGS WITH A NON-COMBUSTIBLE MATERIAL, AND SEAL THEM AIR TIGHT TO PREVENT CONTAMINATION OF THE RESPECTIVE SYSTEMS DURING CONSTRUCTION.
- CONTRACTOR MUST REMOVE AND DISPOSE OF OFFSITE ALL DEMOLISHED WORK IN ACCEPTABLE AND SAFE MANNER AND MUST KEEP ALL NON-WORK AREAS CLEAN AND SAFE.
- ALL EXISTING EQUIPMENT AND CONNECTIONS THAT NEED TO BE TEMPORARILY DEMOLISHED FOR RIGGING AND / OR INSTALLATION MUST BE REINSTALLED AND BROUGHT BACK TO ORIGINAL CONDITIONS PRIOR TO TEMPORARY REMOVAL.
- INSTALL WORK SO AS TO BE READILY ACCESSIBLE FOR OPERATION, MAINTENANCE AND REPAIR. MINOR DEVIATIONS FROM DRAWINGS MAY BE MADE TO ACCOMPLISH THIS, BUT CHANGES WHICH INVOLVE EXTRA COST MUST NOT BE MADE WITHOUT APPROVAL.

MECHANICAL DEMOLITION NOTES:

- THE CONTRACTOR MUST REVIEW THE DRAWINGS AND SPECIFICATIONS FOR DEMOLITION REQUIREMENTS AND LAYOUT HIS WORK IN A COMPATIBLE AND COMPLEMENTARY MANNER. REMOVE ALL EQUIPMENT, DUCTWORK, SUPPORTS, CONTROLS, ACCESSORIES, ETC., AND MECHANICAL ITEMS MADE OBSOLETE BY THESE ALTERATIONS AS SHOWN IN THE MECHANICAL DRAWINGS. ALL ITEMS TO BE REMOVED OR MODIFIED MAY NOT BE SHOWN, HOWEVER, THIS CONTRACTOR MUST REMOVE ANY MECHANICAL WORK AS REQUIRED BY THE CONSTRUCTION OR AS DIRECTED BY THE GOVERNMENT CONTRACTING OFFICER. SURVEY THE AFFECTED AREAS BEFORE SUBMITTING A BID.
- SCHEDULING OF DEMOLITION - COORDINATE SCHEDULING OF MECHANICAL DEMOLITION WORK WITH THE CONTRACTING OFFICER SO AS TO MINIMIZE DISRUPTION OF THE GOVERNMENT'S USE OF THE FACILITIES AND MAINTAIN THE CONSTRUCTION SEQUENCE. SEE DRAWINGS AND SPECIFICATIONS FOR ADDITIONAL INSTRUCTIONS CONCERNING PHASING AND SEQUENCE OF WORK.
- DEMOLISHED MATERIALS - UNLESS SPECIFICALLY REQUESTED BY THE GOVERNMENT, ALL DEMOLISHED MECHANICAL MATERIALS MUST BECOME THE PROPERTY OF THE CONTRACTOR AND MUST BE REMOVED FROM THE SITE AND DISPOSED OF PROPERLY.
- CUTTING AND PATCHING - PERFORM CUTTING AND PATCHING FOR MECHANICAL WORK SO AS TO MINIMIZE DAMAGE TO CEILINGS, FLOORS AND WALLS.
- THESE DRAWINGS ARE COMPILED BY THE ENGINEER FROM THE GOVERNMENT'S AS-BUILT RECORD DRAWINGS AND LIMITED FIELD VERIFICATION OF EXISTING CONDITIONS FOR THE PURPOSE OF INDICATING THE WORK REQUIRED AND ARE BELIEVED TO BE CORRECT. NOTWITHSTANDING, THE CONTRACTOR MUST VERIFY ALL DUCTWORK, EQUIPMENT LOCATIONS, DIMENSIONS AND ALL FIELD CONDITIONS AFFECTING HIS WORK.
- WHERE MECHANICAL SYSTEMS PASS THROUGH THE DEMOLITION AREAS TO SERVE OTHER PORTIONS OF THE PREMISES, THEY MUST REMAIN OR BE SUITABLY RELOCATED AND THE SYSTEM RESTORED TO NORMAL OPERATION. ADVISE THE CONTRACTING OFFICER IMMEDIATELY IF SUCH CONDITIONS ARE UNCOVERED BEFORE PROCEEDING WITH ADDITIONAL WORK.
- PROTECT ALL EXISTING LIFE SAFETY SYSTEMS, FIRE ALARM AND PUBLIC ADDRESS SYSTEMS AND MAINTAIN THEM IN OPERATION THROUGHOUT THE PROGRESS OF THE WORK. NOTIFY THE CONTRACTING OFFICER IN WRITING IF SHUTDOWNS ARE REQUIRED PRIOR TO ANY OUTAGE OF SERVICE. WHERE THE DURATION OF A PROPOSED OUTAGE CANNOT BE TOLERATED BY THE GOVERNMENT, PROVIDE TEMPORARY CONNECTIONS AS REQUIRED MAINTAINING SERVICE.
- SURVEY THE AFFECTED AREAS BEFORE STARTING DEMOLITION AS ALL EXISTING CONDITIONS CANNOT BE COMPLETELY DEPICTED ON THE DRAWINGS AND SOME UNUSUAL CONDITIONS EXIST.
- IF ANY UNUSUAL STRUCTURAL OR ARCHITECTURAL CONDITIONS ARE ENCOUNTERED DURING DEMOLITION, CONTACT THE CONTRACTING OFFICER FOR ASSISTANCE.

DDC SYSTEM INTEGRATION NOTE:

THE ENERGY MANAGEMENT AND CONTROL SYSTEM (EMCS) AT CAMP LEJEUNE IS BY JOHNSON CONTROLS, INC. (JCI) METASYS. INCLUDE ALL CONTROL POINTS, SEQUENCES, PROGRAMMING, AND GRAPHIC UPDATES FOR NEW AND EXISTING HVAC EQUIPMENT MODIFIED UNDER THIS PROJECT TO MATCH THE BUILDING STANDARD FOR A COMPLETE AND OPERABLE SYSTEM.

MECHANICAL ABBREVIATIONS

ABBREVIATION	TERM
ADJ	ADJUSTABLE
AMCA	AIR MOVEMENT AND CONTROL ASSOCIATION
AMP	AMPERE (AMP, AMPS)
ASTM	AMERICAN SOCIETY OF TESTING AND MATERIALS
CFM	CUBIC FEET PER MINUTE
CIP	CAST IN PLACE
CMU	CONCRETE MASONRY UNIT
COP	COEFFICIENT OF PERFORMANCE
DB	DRY BULB
DEG OR °	DEGREE
EA	EXHAUST AIR
EG	EXHAUST GRILLE
EAT	ENTERING AIR TEMPERATURE
ECM	ELECTRONICALLY COMMUTATED MOTOR
EER	ENERGY EFFICIENCY RATIO
ESP	EXTERNAL STATIC PRESSURE
F	FAN
°F	FAHRENHEIT
FLA	FULL LOAD AMPS
FT	FEET
HC	HOT WATER COIL
HGT OR H	HEIGHT
HP	HORSEPOWER
HR	HOUR(S)
IN.	INCH
IN.-WG	INCHES WATER GAUGE
KW	KILOWATT
LAT	LEAVING AIR TEMPERATURE
LBS	POUNDS
L	LOUVER
MAX	MAXIMUM
MBH	1000 BTUH
MCA	MINIMUM CIRCUIT AMPACITY
MCWB	MEAN COINCIDENT WET BULB
MIN.	MINIMUM
MOCP	MAXIMUM OVER CURRENT PROTECTION
NEMA	NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
OZ	OUNCE
OA	OUTSIDE AIR
%	PERCENT
RA	RETURN AIR
RG	RETURN GRILLE
RPM	REVOLUTIONS PER MINUTE
RTU	ROOF TOP UNIT
SA	SUPPLY AIR
SF	SQUARE-FEET
SG	SUPPLY GRILLE
SQ	SQUARE
TG	TRANSFER GRILLE
TYP	TYPICAL
UH	UNIT HEATER
V/PH/Hz	VOLT/PHASE/HERTZ
VTR	VENT THROUGH ROOF
W	WIDTH
WB	WET BULB

NOTE: ALL ABBREVIATIONS MAY NOT BE USED IN PROJECT.

MECHANICAL LEGEND

	CEILING EXHAUST AIR GRILLE
	CEILING RETURN AIR / TRANSFER AIR GRILLE
	CEILING SUPPLY AIR DIFFUSER / GRILLE
	DIFFUSER / REGISTER / GRILLE TAG
	
	EXTENT OF DEMOLITION
	HUMIDISTAT / HUMIDITY SENSOR
	INDICATES TO DEMOLISH
	MANUAL VOLUME DAMPER
	MOTORIZED DAMPER
	POINT OF CONNECTION
	RETURN, EXHAUST OR TRANSFER AIR FLOW
	SUPPLY AIR FLOW
	THERMOSTAT / TEMPERATURE SENSOR
	T-STAT / HUMIDISTAT OR TEMPHUMIDITY SENSOR

NOTE: ALL ITEMS LISTED MAY NOT BE USED IN THIS PROJECT.

REVISIONS:

FINAL
02-24-2026

G001

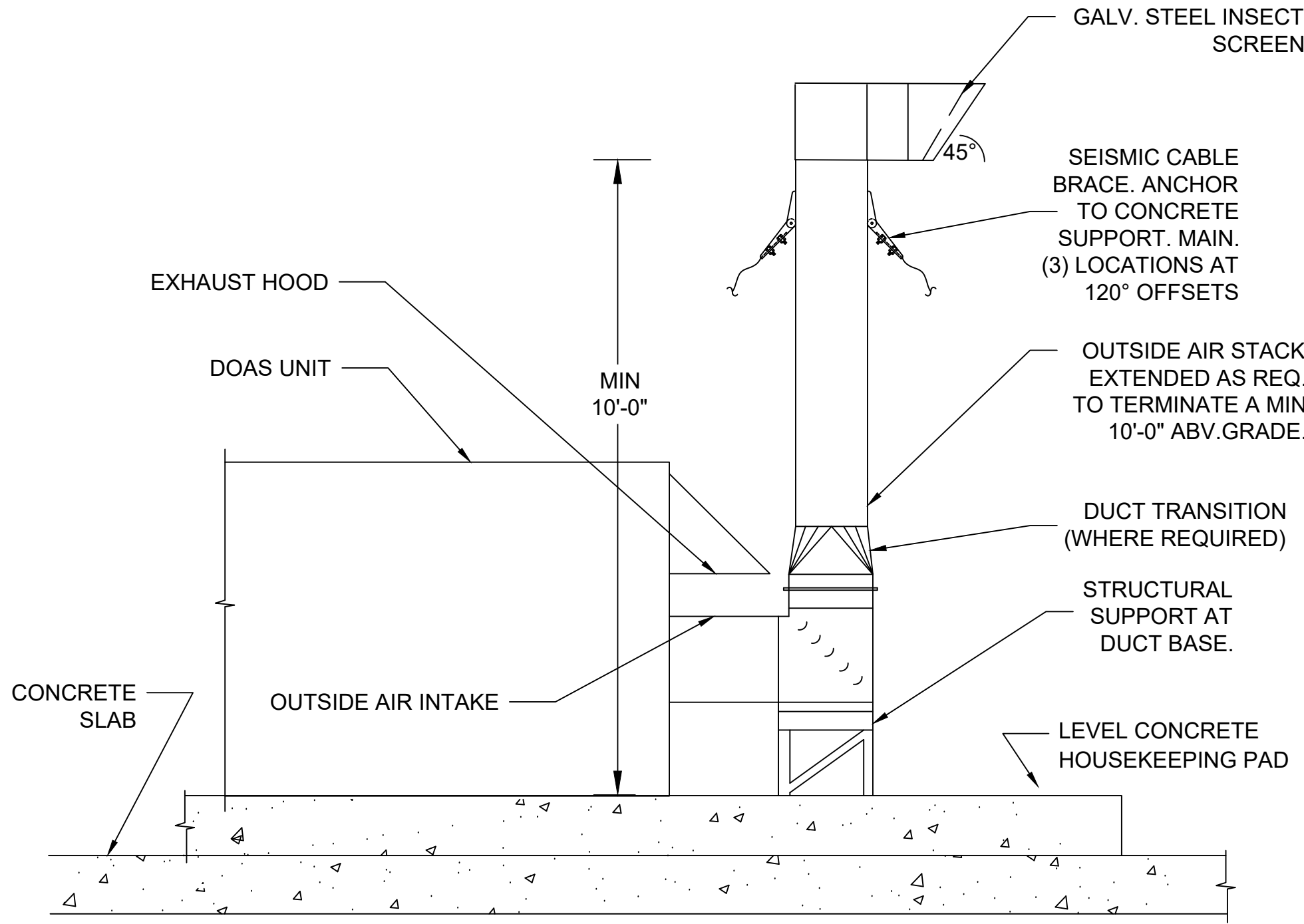
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND
MARINE CORPS BASE
CAMP LEJEUNE, NORTH CAROLINA

GEIGER QUAD G550
DOAS REPLACEMENT

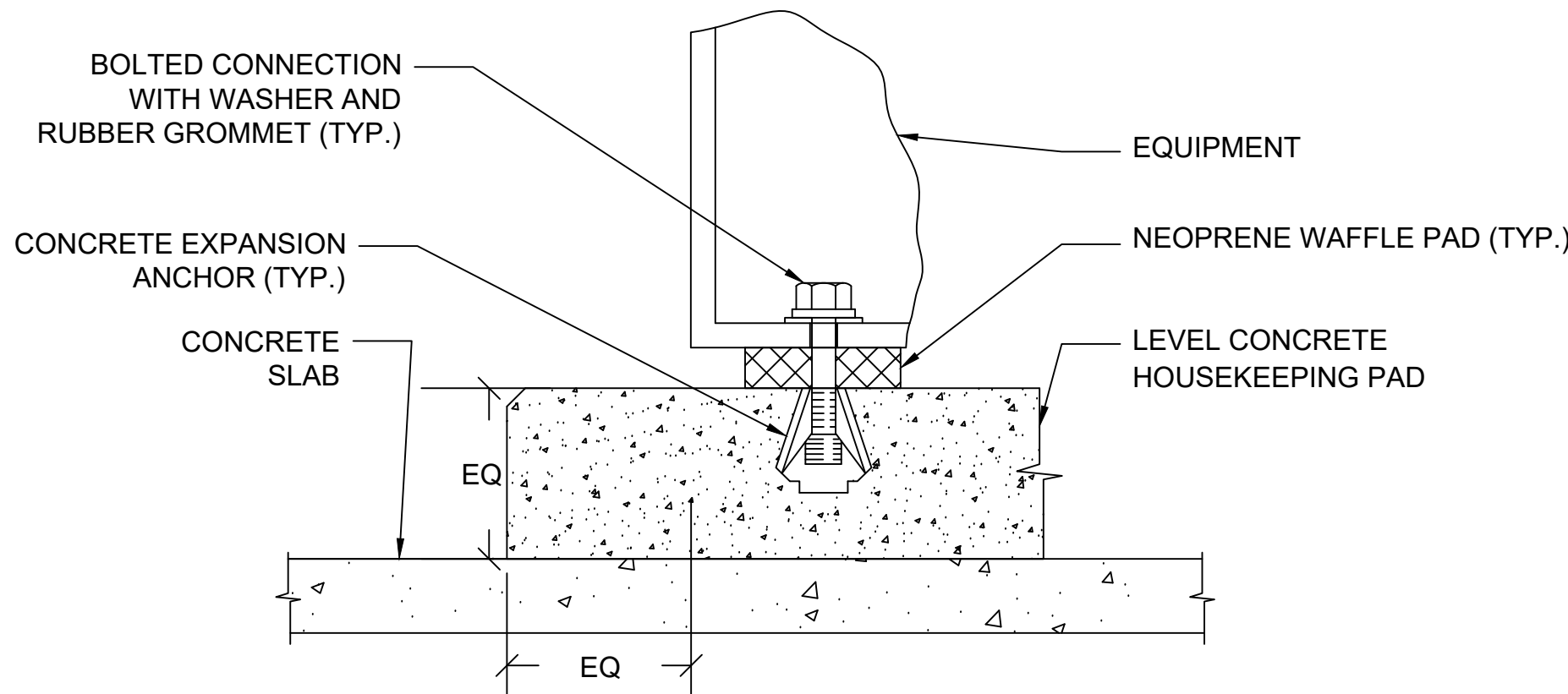
GENERAL
NOTES AND ABBREVIATIONS

DES. CJH	DR. CJH	CHK. CJM	SUBMITTED BY: CJM	DESIGN DIR.C. Moos, PE	APPROVED: PWO OR OICC	DATE	SIZE	CODE	IDENT. NO	NAVFAC DRAWING NO.
							E1		80091	60042369
										CONST. CONTR. N40085-25-B-0080
SATISFACTORY TO:						DATE	SCALE:	NOTED	SPEC. 05-25-0080	SHEET 2 OF 8

REVISIONS:	



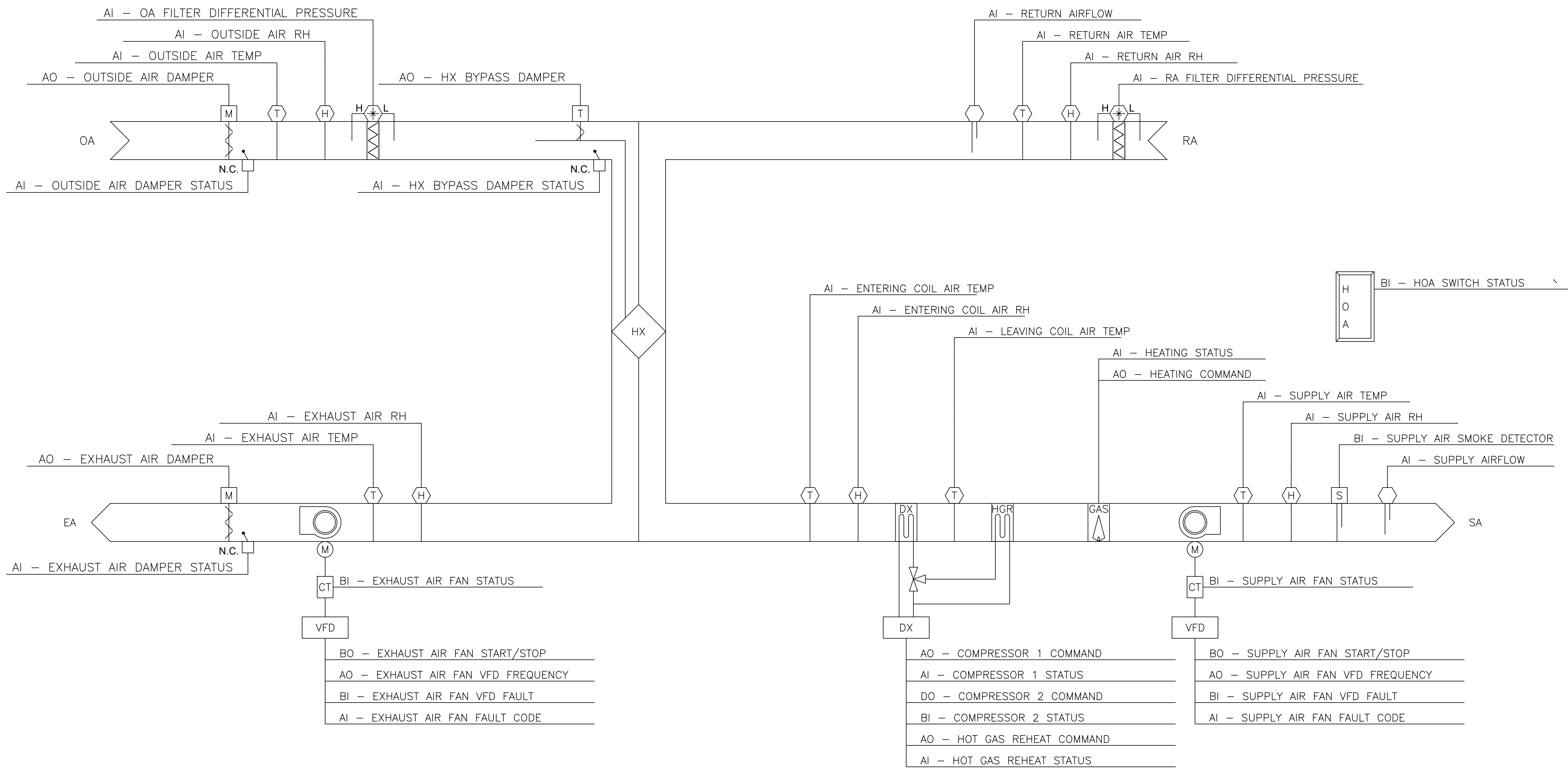
1 DOAS VERTICAL DUCT EXTENSION
NOT TO SCALE



2 EQUIPMENT MOUNTING DETAIL
NOT TO SCALE

DEDICATED OUTSIDE AIR UNIT SCHEDULE																	
MARK	LOCATION	OUTDOOR AMBIENT		SUPPLY AIR (DB/WB °F)	RETURN AIR (DB/WB °F)	SUPPLY FAN DATA		EXHAUST FAN DATA		HEATING			ELECTRICAL			NOTES	ACCESSORIES
		SUMMER (DB/WB °F)	WINTER (DB °F)			OUTSIDE AIR (CFM)	ESP (IN WG)	EXHAUST AIR (CFM)	ESP (IN WG)	FUEL TYPE	EAT (°F)	INPUT (MBH)	VOLTS/PH/HZ	MCA	MOCP		
DOAS-1,2,3,4	G551, G551A, G551B, G551C	84/78.5	22	72.0/59.0	72.0/61.4	5,100	2.0	4,900	1.5	NATURAL GAS	40	200	460/3/60	64.1	80	1,2,4,5	A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P
DOAS-5	G550	84/78.5	22	72.0/59.0	-	770	0.5	-	-	HEAT PUMP	22	-	460/3/60	13.2	20	1,2,3,5	A,B,C,D,E,G,H,I,K,L,M,O,P
NOTES: 1) REFER TO SPECIFICATION SECTION 23 73 33 - HEATING, VENTILATION, AND COOLING SYSTEM FOR FURTHER REQUIREMENTS. 2) EFFICIENCY RATED IN ACCORDANCE WITH ANSI/AHRI STANDARD 340/360. 3) HOT GAS REHEAT AND HEAT PUMP HEATING IS INTEGRAL TO UNIT. UTILIZE EXISTING AUXILIARY ELECTRIC HEATING. 4) HEATING SHALL BE DONE VIA NATURAL GAS. DX SHALL ONLY BE USED FOR COOLING AND DEHUMIDIFICATION. 5) SETPOINTS SHALL BE ADJUSTABLE WITHOUT THE USE OF ANY ADDITIONAL ACCESSORIES NOT PERMANENTLY INSTALLED ON THE UNIT.																	
ACCESSORIES: A) CONDENSER, EVAPORATOR AND REHEAT COILS TO BE COATED FOR EXPOSURE TO ASTM B117-90 3,000 HOUR SALT SPRAY RESISTANCE TEST WITH NO DEGRADATION. B) SINGLE POINT ELECTRICAL CONNECTION WITH STAINLESS STEEL NEMA 4X DISCONNECT. C) MODULATING HOT GAS REHEAT. D) MERV 8 PRE-FILTER AND MERV 13 FILTER ON OUTDOOR AND RETURN AIR. PROVIDE 1 EXTRA COMPLETE SET OF FILTERS TO THE GOVERNMENT. E) HAIL GUARD. F) ERV FACE/BYPASS DAMPER WITH MODULATING ACTUATOR. ELECTRIC PREHEAT MAY BE PROVIDED AS AN ALTERNATE MEANS OF FROST CONTROL. G) 120V GFCI CONVENIENCE OUTLET. H) AIR COOLED VARIABLE SPEED HEAD PRESSURE CONTROL. I) PHASE AND VOLTAGE MONITOR. J) EXHAUST AIR GRAVITY DAMPER. K) MANUFACTURER DDC CONTROLS WITH BACNET MS/TP INTERFACE. L) OUTSIDE AIR DAMPER WITH 2-POSITION ACTUATOR. DAMPERS SHALL BE SEACOAST DESIGN AND LOW LEAKAGE WITH RUBBER EDGES. M) 304 STAINLESS STEEL CONDENSATE DRAIN PAN. N) PLATE HEAT EXCHANGER. O) FULL 5 YEAR PARTS, LABOR, AND REFRIGERANT WARRANTY. P) FACTORY STARTUP. REPORT TO BE PROVIDED TO THE GOVERNMENT 5 DAYS AFTER STARTUP.																	

FINAL			M501		
02-24-2026					
		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND			
		MARINE CORPS BASE			
		CAMP LEJEUNE, NORTH CAROLINA			
DES. CJH		GEIGER QUAD G550 DOAS REPLACEMENT			
DR. CJH					
CHK. CJM					
SUBMITTED BY: CJM					
DESIGN DIR.C. Moos, PE		MECHANICAL DETAILS AND SCHEDULE			
APPROVED: PWO OR OICC DATE					
		SIZE	CODE IDENT. NO	NAVFAC DRAWING NO.	
		E1	80091	60042371	
				CONST. CONTR. N40085-25-B-0080	
SATISFACTORY TO: DATE		SCALE: NOTED		SPEC. 05-25-0080 SHEET 4 OF 8	



1 BARRACKS DOAS DDC CONTROL DIAGRAM
NOT TO SCALE

REVISIONS:

KEYNOTES:

1 ELECTRIC PREHEAT IS ACCEPTABLE IN LIEU OF FACE AND BYPASS DAMPER. IF ELECTRIC PREHEAT IS PROVIDED, PROVIDE COMMAND AND STATUS POINT. CONTRACTOR SHALL MAKE ALL NECESSARY MODIFICATIONS REQUIRED IF ELECTRICAL REQUIREMENTS EXCEED CURRENT INSTALLED CAPACITY.

FINAL
02-24-2026

M801

DES. CJH		MARINE CORPS BASE	
DR. CJH		CAMP LEJEUNE, NORTH CAROLINA	
CHK. CJM		GEIGER QUAD G550	
SUBMITTED BY: CJM		DOAS REPLACEMENT	
DESIGN DIR.C. Moos, PE		MECHANICAL BARRACKS	
APPROVED: PWO OR OICC		DOAS DDC DIAGRAM	
DATE		SIZE	CODE IDENT. NO.
DATE		E1	80091
SATSFACTORY TO:		SCALE: NOTED	NAVAFAC DRAWING NO. 60042372
DATE			CONST. CONTR. N40085-25-B-0080
			SHEET 5 OF 8

REVISIONS:	

BARRACKS DOAS SEQUENCE OF OPERATIONS		
A. GENERAL	EAD CLOSED.	
1. PROGRAMS AND SEQUENCES INDICATED AND IMPLIED HEREIN SHALL BE PROVIDED.	C. AIRFLOW CONTROL:	3. GENERATE AN ALARM AT THE BAS IF THE SUPPLY OR EXHAUST AIRFLOW IS ±10% (ADJ.) FROM SETPOINT FOR MORE THAN 30 MINUTES (ADJ.).
2. PROVIDE ALL I/O POINTS INDICATED ON THE CONTROL DIAGRAM AND AS REQUIRED TO ACCOMPLISH THE SEQUENCE OF OPERATION.	1. WHEN FAN STATUS INDICATES THE FAN HAS STARTED, THE FANS SHALL MODULATE TO MAINTAIN AIRFLOW SETPOINTS USING PID CONTROL LOOP.	4. GENERATE AN ALARM AT THE BAS IF THE SUPPLY AIR DEWPOINT EXCEEDS +5°F (ADJ.) FROM SETPOINT FOR MORE THAN 30 MINUTES (ADJ.).
3. THE VFD/MCC HOA SWITCH SHALL BE KEPT IN THE AUTO POSITION. HAND AND OFF POSITION SHALL BE UTILIZED FOR MAINTENANCE PURPOSES ONLY.	D. TEMPERATURE AND HUMIDITY CONTROL:	5. GENERATE AN ALARM AT THE BAS IF A DAMPER STATUS DOES NOT MATCH THE COMMAND.
4. ALL POINTS SHALL BE ABLE TO INTEGRATE TO ALL TRENDS TOTALIZATIONS, GRAPHICS PAGES ETC., AS APPLICABLE.	1. HEATING:	6. GENERATE AN ALARM AT THE BAS UPON RECEIVING A SMOKE DETECTOR SIGNAL.
5. ALL SETTINGS AND SETPOINTS SHALL BE USER ADJUSTABLE WITHOUT REQUIRING A MODIFICATION TO THE PROGRAMMING LOGIC CODE.	a. IF THE SUPPLY AIR TEMPERATURE IS LESS THAN THE SUPPLY AIR TEMPERATURE SETPOINT AND THE SUPPLY AIR DEWPOINT IS LESS THAN 50°F, THE UNIT WILL ENABLE THE NATURAL GAS HEATER AND MODULATE CAPACITY TO MAINTAIN THE SUPPLY AIR TEMPERATURE SETPOINT.	7. GENERATE AN ALARM AT THE BAS IF A COMPRESSOR STATUS DOES NOT MATCH THE COMMAND.
6. ALL ALARMS AS INDICATED ON THE DIAGRAMS IN THE SEQUENCE OF OPERATION SHALL BE GENERATED TO THE APPROPRIATE SERVER/WORKSTATIONS.	b. DX HEATING OR DX COOLING WITH REHEAT SHALL NOT BE UTILIZED IN HEATING MODE.	8. GENERATE AN ALARM AT THE BAS IF THE UNIT IS IN HAND.
B. SETPOINTS:	2. DEHUMIDIFICATION:	F. SAFETY INTERLOCKS
1. SUPPLY AIR TEMPERATURE: 72.0°F	a. IF THE SUPPLY AIR DEWPOINT IS GREATER THAN 50°F, THE UNIT WILL ENABLE THE DX COOLING AND REHEAT AND MODULATE CAPACITY TO MAINTAIN THE SUPPLY AIR TEMPERATURE AND DEWPOINT SETPOINT.	1. PROVIDE ANTI-TERRORISM FORCE PROTECTION (ATFP) EMERGENCY SHUTDOWN BOTH HARDWIRED IN THE EQUIPMENT SAFETY CIRCUIT AND IN SOFTWARE FROM THE BAS. THE SHUTDOWN ACTIVATION SHALL LATCH AND REQUIRE MANUAL RESET. UPON ACTIVATION, THE UNIT SHALL:
2. SUPPLY AIR DEWPOINT: 50.0°F	3. COOLING:	a. OUTSIDE AIR DAMPER AND EXHAUST AIR DAMPER SHALL CLOSE WITHIN 30 SECONDS.
C. START/STOP/SCHEDULING:	a. IF THE SUPPLY AIR TEMPERATURE IS GREATER THAN THE SUPPLY AIR TEMPERATURE SETPOINT AND THE SUPPLY AIR DEWPOINT IS LESS THAN 50°F, THE UNIT WILL ENABLE THE DX COOLING AND REHEAT AND MODULATE CAPACITY TO MAINTAIN THE SUPPLY AIR TEMPERATURE SETPOINT.	b. EXHAUST FAN AND SUPPLY AIR FAN SHALL STOP.
1. THE DOAS OPERATES CONTINUOUSLY WITH REMOTE CAPABILITY FOR START/STOP/SCHEDULING FUNCTIONS.	b. NATURAL GAS HEATER SHALL BE ENABLED IF REHEAT CANNOT MAINTAIN THE SUPPLY AIR TEMPERATURE SETPOINT.	2. HX BYPASS DAMPER SHALL MODULATE OPEN TO PREVENT THE ENTERING COIL TEMPERATURE FROM FALLING BELOW THE FROST THRESHOLD SETPOINT AS REQUIRED BY THE HX MANUFACTURER (26°F ADJ.)
a. PROVIDE INDIVIDUAL, USER DEFINABLE SCHEDULES FOR OCCUPIED AND UNOCCUPIED TIME PERIODS.	E. ALARMS:	3. INTEGRATE HARDWIRED SMOKE DETECTOR IN THE EQUIPMENT SAFETY CIRCUIT AND IN SOFTWARE FROM THE BAS. UPON ACTIVATION OF THE SMOKE DETECTOR, THE UNIT SHALL:
b. OCCUPANCY HOURS WILL BE CONTINUOUS.	1. IF AFTER 60 SECOND DELAY (ADJ.), THE SUPPLY OR EXHAUST FAN DOES NOT PROVIDE STATUS OF ON, THE FANS WILL BE SHUT DOWN, THE OAD WILL CLOSE, THE EAD WILL CLOSE, AND AN ALARM WILL BE GENERATED. THE SYSTEM WILL RESTART AUTOMATICALLY WHEN FAULT CONDITIONS HAVE CLEARED.	a. OUTSIDE AIR DAMPER AND EXHAUST AIR DAMPER SHALL CLOSE.
2. WHEN AHU IS STARTED (MANUALLY OR AUTOMATICALLY):	2. GENERATE AN ALARM AT THE BAS IF THE DISCHARGE AIR TEMPERATURE EXCEEDS ±5°F (ADJ.) FROM SETPOINT FOR MORE THAN 30 MINUTES (ADJ.).	b. EXHAUST FAN AND SUPPLY AIR FAN SHALL STOP.
a. THE OUTSIDE AIR DAMPER (OAD) AND EXHAUST AIR DAMPERS (EAD) SHALL OPEN TO 100%. FANS SHALL BE ENERGIZED AFTER THEIR RESPECTIVE DAMPERS ARE PROVED OPEN.		
3. WHEN DOAS IS STOPPED (EITHER MANUALLY, AUTOMATICALLY DURING UNOCCUPIED MODE, OR FROM SAFETY FUNCTION):		
a. THE SUPPLY AND EXHAUST FAN SHALL BE DE-ENERGIZED, OAD CLOSED, AND		

1

BARRACKS DOAS SEQUENCE OF OPERATION

NOT TO SCALE

BARRACKS DOAS UNIT POINTS LIST												
POINT NAME	HARDWARE POINTS				SOFTWARE POINTS						SHOW ON GRAPHIC	
	AI	AO	BI	BO	AV	BV	LOOP	SCHED	TREND	ALARM		
SUPPLY AIR FAN STATUS			X						X		X	
SUPPLY AIR FAN START/STOP				X					X		X	
SUPPLY AIR FAN VFD FREQUENCY		X							X		X	
SUPPLY AIR FAN VFD FREQUENCY SETPOINT					X				X		X	
SUPPLY AIR FAN VFD FAULT			X						X	X	X	
SUPPLY AIR FAN FAULT CODE	X								X		X	
HOA STATUS SWITCH			X						X		X	
SUPPLY AIR SMOKE DETECTOR			X						X		X	
SUPPLY AIR TEMPERATURE	X								X		X	
SUPPLY AIR TEMPERATURE SETPOINT					X				X		X	
SUPPLY AIR RH	X								X		X	
SUPPLY AIR DEWPOINT					X				X		X	
SUPPLY AIR DEWPOINT SETPOINT					X				X		X	
SUPPLY AIRFLOW	X								X		X	
SUPPLY AIRFLOW SETPOINT					X				X		X	
HEATING STATUS			X						X		X	
HEATING COMMAND				X					X		X	
OUTSIDE AIR DAMPER		X							X		X	
OUTSIDE AIR DAMPER POSITION	X								X		X	
OUTSIDE AIR TEMPERATURE	X								X		X	
OUTSIDE AIR RH	X								X		X	
OUTSIDE AIR FILTER DIFFERENTIAL PRESSURE	X								X		X	
HX BYPASS DAMPER		X							X		X	
HX BYPASS DAMPER STATUS	X								X		X	
RETURN AIR TEMPERATURE	X								X		X	
RETURN AIR RH	X								X		X	
RETURN AIRFLOW	X								X		X	
RETURN AIR FILTER DIFFERENTIAL PRESSURE	X								X		X	
ENTERING COIL AIR TEMPERATURE	X								X		X	
ENTERING COIL RH	X								X		X	
LEAVING COIL AIR TEMPERATURE	X								X		X	
EXHAUST AIR TEMPERATURE	X								X		X	
EXHAUST AIR RH	X								X		X	
EXHAUST AIR DAMPER		X							X		X	
EXHAUST AIR DAMPER STATUS	X								X		X	

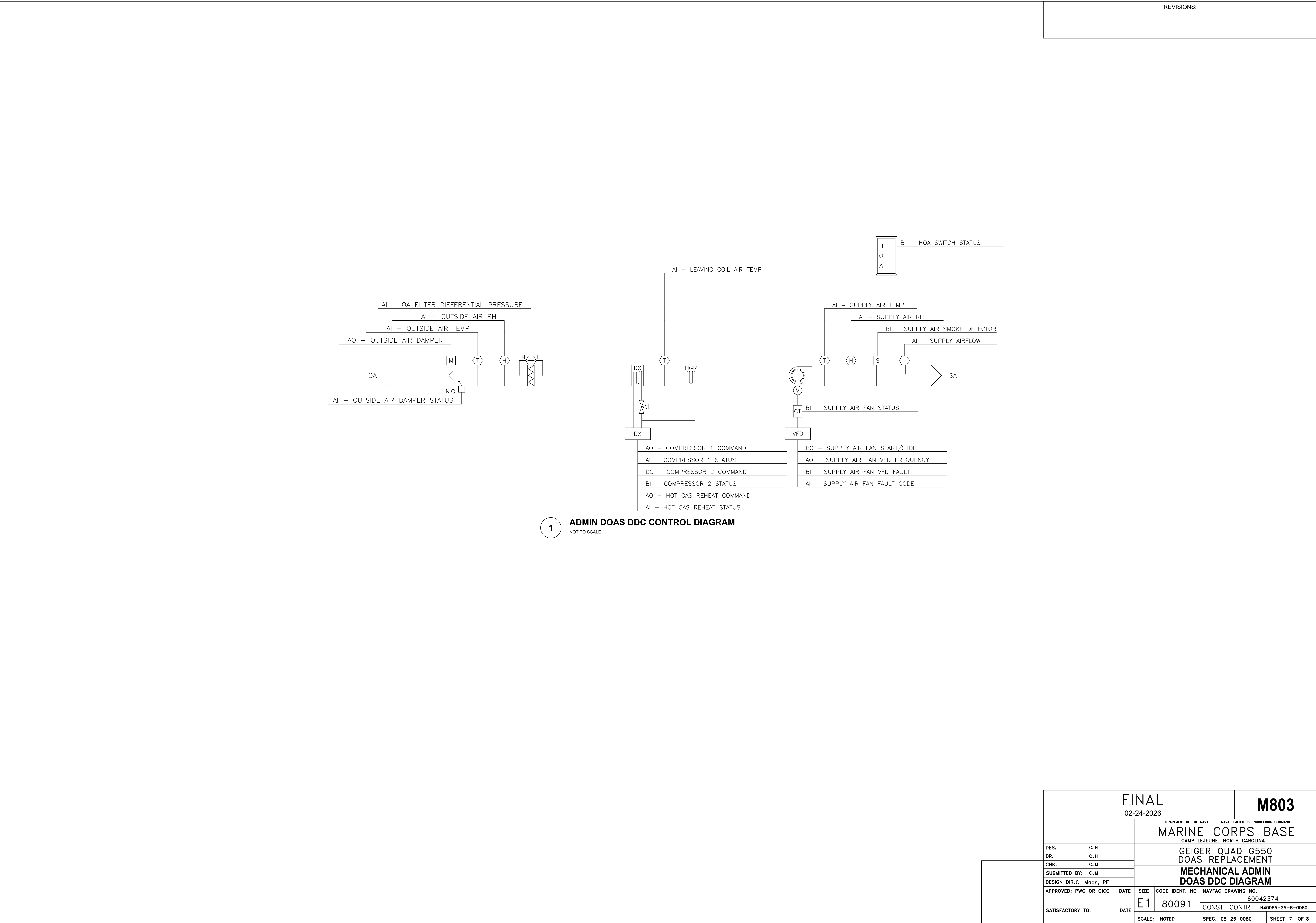
2

BARRACKS DOAS POINTS LIST

NOT TO SCALE

BARRACKS DOAS UNIT POINTS LIST												
POINT NAME	HARDWARE POINTS				SOFTWARE POINTS						SHOW ON GRAPHIC	
	AI	AO	BI	BO	AV	BV	LOOP	SCHED	TREND	ALARM		
UNIT ENABLE			X						X		X	
EXHAUST AIR FAN STATUS			X						X		X	
EXHAUST AIR FAN START/STOP				X					X		X	
EXHAUST AIR FAN VFD FREQUENCY		X							X		X	
EXHAUST AIR FAN FREQUENCY SETPOINT					X				X		X	
EXHAUST AIR FAN VFD FAULT			X						X	X	X	
EXHAUST AIR FAN FAULT CODE	X								X		X	
COMPRESSOR 1 COMMAND		X							X		X	
COMPRESSOR 1 STATUS	X								X		X	
COMPRESSOR 2 COMMAND				X					X		X	
COMPRESSOR 2 STATUS			X						X		X	
HOT GAS REHEAT COMMAND		X							X		X	
HOT GAS REHEAT STATUS	X								X		X	
SUPPLY AIR FAN ALARM									X		X	
EXHAUST AIR FAN ALARM									X		X	
HIGH SUPPLY AIR TEMPERATURE									X	X	X	
LOW SUPPLY AIR TEMPERATURE									X	X	X	
HIGH SUPPLY AIR DEWPOINT									X	X	X	
HIGH RETURN AIR TEMPERATURE									X	X	X	
LOW RETURN AIR TEMPERATURE									X	X	X	
HIGH RETURN AIR RH									X	X	X	
COMPRESSOR 1 ALARM									X	X	X	
COMPRESSOR 2 ALARM									X	X	X	
OUTSIDE AIR DAMPER ALARM									X	X	X	
EXHAUST AIR DAMPER ALARM									X	X	X	
LOW SUPPLY AIRFLOW									X	X	X	
HIGH SUPPLY AIRFLOW									X	X	X	
LOW RETURN AIRFLOW									X	X	X	
HIGH RETURN AIRFLOW									X	X	X	
HIGH OA FILTER DIFFERENTIAL PRESSURE									X	X	X	
HIGH RA FILTER DIFFERENTIAL PRESSURE									X	X	X	
HX BYPASS DAMPER ALARM									X	X	X	

FINAL		M802	
02-24-2026			
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND			
MARINE CORPS BASE			
CAMP LEJEUNE, NORTH CAROLINA			
DES. CJH		GEIGER QUAD G550	
DR. CJH		DOAS REPLACEMENT	
CHK. CJM			
SUBMITTED BY: CJM		MECHANICAL BARRACKS	
DESIGN DIR.C. Moos, PE		DOAS DDC POINTS	
APPROVED: PWO OR OICC	DATE	SIZE	CODE IDENT. NO
		E1	80091
SATISFACTORY TO:		DATE	NAVFAC DRAWING NO.
			60042373
		SCALE: NOTED	CONST. CONTR. N40085-25-B-0080
			SHEET 6 OF 8



REVISIONS:	

ADMIN DOAS SEQUENCE OF OPERATIONS		
A.GENERAL	1. WHEN FAN STATUS INDICATES THE FAN HAS STARTED, THE FAN SHALL MODULATE TO MAINTAIN AIRFLOW SETPOINT USING PID CONTROL LOOP.	SETPOINT FOR MORE THAN 30 MINUTES (ADJ.).
1. PROGRAMS AND SEQUENCES INDICATED AND IMPLIED HEREIN SHALL BE PROVIDED.	D.TEMPERATURE AND HUMIDITY CONTROL:	4. GENERATE AN ALARM AT THE BAS IF THE SUPPLY AIR DEWPOINT EXCEEDS +5°F (ADJ.) FROM SETPOINT FOR MORE THAN 30 MINUTES (ADJ.).
2. PROVIDE ALL I/O POINTS INDICATED ON THE CONTROL DIAGRAM AND AS REQUIRED TO ACCOMPLISH THE SEQUENCE OF OPERATION.	1. HEATING:	5. GENERATE AN ALARM AT THE BAS IF A DAMPER STATUS DOES NOT MATCH THE COMMAND.
3. THE VFD/MCC HOA SWITCH SHALL BE KEPT IN THE AUTO POSITION. HAND AND OFF POSITION SHALL BE UTILIZED FOR MAINTENANCE PURPOSES ONLY.	a. IF THE SUPPLY AIR TEMPERATURE IS LESS THAN THE SUPPLY AIR TEMPERATURE SETPOINT AND THE SUPPLY AIR DEWPOINT IS LESS THAN 50°F, THE UNIT WILL ENABLE THE DX HEATING MODULATE CAPACITY TO MAINTAIN THE SUPPLY AIR TEMPERATURE SETPOINT.	6. GENERATE AN ALARM AT THE BAS UPON RECEIVING A SMOKE DETECTOR SIGNAL.
4. ALL POINTS SHALL BE ABLE TO INTEGRATE TO ALL TRENDS TOTALIZATIONS, GRAPHICS PAGES ETC., AS APPLICABLE.	b. DURING DEFROST OR IF THE SUPPLY AIR SETPOINT CANNOT BE MAINTAINED, ENABLE THE EXISTING DUCT HEATER.	7. GENERATE AN ALARM AT THE BAS IF A COMPRESSOR STATUS DOES NOT MATCH THE COMMAND.
5. ALL SETTINGS AND SETPOINTS SHALL BE USER ADJUSTABLE WITHOUT REQUIRING A MODIFICATION TO THE PROGRAMMING LOGIC CODE.	2. DEHUMIDIFICATION:	8. GENERATE AN ALARM AT THE BAS IF THE UNIT IS IN HAND.
6. ALL ALARMS AS INDICATED ON THE DIAGRAMS IN THE SEQUENCE OF OPERATION SHALL BE GENERATED TO THE APPROPRIATE SERVER/WORKSTATIONS.	a. IF THE SUPPLY AIR DEWPOINT IS GREATER THAN 50°F, THE UNIT WILL ENABLE THE DX COOLING AND REHEAT AND MODULATE CAPACITY TO MAINTAIN THE SUPPLY AIR TEMPERATURE AND DEWPOINT SETPOINT.	F.SAFETY INTERLOCKS
B.SETPOINTS:	b. IF THE SUPPLY AIR TEMPERATURE SETPOINT CANNOT BE MAINTAINED, ENABLE THE EXISTING DUCT HEATER.	1. PROVIDE ANTI-TERRORISM FORCE PROTECTION (ATFP) EMERGENCY SHUTDOWN BOTH HARDWIRED IN THE EQUIPMENT SAFETY CIRCUIT AND IN SOFTWARE FROM THE BAS. THE SHUTDOWN ACTIVATION SHALL LATCH AND REQUIRE MANUAL RESET. UPON ACTIVATION, THE UNIT SHALL:
1. SUPPLY AIR TEMPERATURE: 72.0°F	3. COOLING:	a. OUTSIDE AIR DAMPER SHALL CLOSE WITHIN 30 SECONDS.
2. SUPPLY AIR DEWPOINT: 50.0°F	a. IF THE SUPPLY AIR TEMPERATURE IS GREATER THAN THE SUPPLY AIR TEMPERATURE SETPOINT AND THE SUPPLY AIR DEWPOINT IS LESS THAN 50°F, THE UNIT WILL ENABLE THE DX COOLING AND REHEAT AND MODULATE CAPACITY TO MAINTAIN THE SUPPLY AIR TEMPERATURE SETPOINT.	b. SUPPLY AIR FAN SHALL STOP.
C.START/STOP/SCHEDULING:	b. IF THE SUPPLY AIR TEMPERATURE SETPOINT CANNOT BE MAINTAINED, ENABLE THE EXISTING DUCT HEATER.	2. INTEGRATE HARDWIRED SMOKE DETECTOR IN THE EQUIPMENT SAFETY CIRCUIT AND IN SOFTWARE FROM THE BAS. UPON ACTIVATION OF THE SMOKE DETECTOR, THE UNIT SHALL:
1. THE DOAS OPERATES CONTINUOUSLY WITH REMOTE CAPABILITY FOR START/STOP/SCHEDULING FUNCTIONS.	E.ALARMS:	a. OUTSIDE AIR DAMPER SHALL CLOSE.
a. PROVIDE INDIVIDUAL, USER DEFINABLE SCHEDULES FOR OCCUPIED AND UNOCCUPIED TIME PERIODS.	1. IF AFTER 60 SECOND DELAY (ADJ.), THE SUPPLY FAN DOES NOT PROVIDE STATUS OF ON, THE FAN WILL BE SHUT DOWN, THE OAD WILL CLOSE, AND AN ALARM WILL BE GENERATED. THE SYSTEM WILL RESTART AUTOMATICALLY WHEN FAULT CONDITIONS HAVE CLEARED.	b. SUPPLY AIR FAN SHALL STOP.
b. OCCUPANCY HOURS WILL BE 0600–1800.	2. GENERATE AN ALARM AT THE BAS IF THE DISCHARGE AIR TEMPERATURE EXCEEDS ±5°F (ADJ.) FROM SETPOINT FOR MORE THAN 30 MINUTES (ADJ.).	c. 3. INTERLOCK SUPPLY FAN WITH EXISTING DUCT HEATERS. DUCT HEATER MUST BE ENERGIZE UNLESS DESIGN SUPPLY AIRFLOW IS PROVEN.
2. WHEN AHU IS STARTED (MANUALLY OR AUTOMATICALLY):	3. GENERATE AN ALARM AT THE BAS IF THE SUPPLY AIRFLOW IS ±10% (ADJ.) FROM	
a. THE OUTSIDE AIR DAMPER (OAD) SHALL OPEN TO 100%. FAN SHALL BE ENERGIZED AFTER OAD IS PROVED OPEN.		
3. WHEN DOAS IS STOPPED (EITHER MANUALLY, AUTOMATICALLY DURING UNOCCUPIED MODE, OR FROM SAFETY FUNCTION):		
a. THE SUPPLY FAN SHALL BE DE-ENERGIZED AND OAD CLOSED.		
C.AIRFLOW CONTROL:		

1

ADMIN DOAS SEQUENCE OF OPERATION

NOT TO SCALE

ADMIN DOAS UNIT POINTS LIST												
POINT NAME	HARDWARE POINTS				SOFTWARE POINTS						SHOW ON GRAPHIC	
	AI	AO	BI	BO	AV	BV	LOOP	SCHED	TREND	ALARM		
SUPPLY AIR FAN STATUS			X						X		X	
SUPPLY AIR FAN START/STOP				X					X		X	
SUPPLY AIR FAN VFD FREQUENCY	X								X		X	
SUPPLY AIR FAN VFD FREQUENCY SETPOINT					X				X		X	
SUPPLY AIR FAN VFD FAULT			X						X	X	X	
SUPPLY AIR FAN FAULT CODE	X								X		X	
HOA STATUS SWITCH			X						X		X	
SUPPLY AIR SMOKE DETECTOR			X						X		X	
SUPPLY AIR TEMPERATURE	X								X		X	
SUPPLY AIR TEMPERATURE SETPOINT					X				X		X	
SUPPLY AIR RH	X								X		X	
SUPPLY AIR DEWPOINT					X				X		X	
SUPPLY AIR DEWPOINT SETPOINT					X				X		X	
SUPPLY AIRFLOW	X								X		X	
SUPPLY AIRFLOW SETPOINT					X				X		X	
OUTSIDE AIR DAMPER		X							X		X	
OUTSIDE AIR DAMPER POSITION	X								X		X	
OUTSIDE AIR TEMPERATURE	X								X		X	
OUTSIDE AIR RH	X								X		X	
OUTSIDE AIR FILTER DIFFERENTIAL PRESSURE	X								X		X	
LEAVING COIL AIR TEMPERATURE	X								X		X	

ADMIN DOAS UNIT POINTS LIST												
POINT NAME	HARDWARE POINTS				SOFTWARE POINTS						SHOW ON GRAPHIC	
	AI	AO	BI	BO	AV	BV	LOOP	SCHED	TREND	ALARM		
UNIT ENABLE			X						X		X	
COMPRESSOR 1 COMMAND		X							X		X	
COMPRESSOR 1 STATUS	X								X		X	
COMPRESSOR 2 COMMAND				X					X		X	
COMPRESSOR 2 STATUS			X						X		X	
HOT GAS REHEAT COMMAND		X							X		X	
HOT GAS REHEAT STATUS	X								X		X	
SUPPLY AIR FAN ALARM									X		X	
HIGH SUPPLY AIR TEMPERATURE									X	X	X	
LOW SUPPLY AIR TEMPERATURE									X	X	X	
HIGH SUPPLY AIR DEWPOINT									X	X	X	
COMPRESSOR 1 ALARM									X	X	X	
COMPRESSOR 2 ALARM									X	X	X	
OUTSIDE AIR DAMPER ALARM									X	X	X	
LOW SUPPLY AIRFLOW									X	X	X	
HIGH SUPPLY AIRFLOW									X	X	X	
HIGH OA FILTER DIFFERENTIAL PRESSURE									X	X	X	

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ADMIN DOAS POINTS LIST

NOT TO SCALE

FINAL				M804	
02-24-2026					
		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND			
		MARINE CORPS BASE			
		CAMP LEJEUNE, NORTH CAROLINA			
DES. CJH		GEIGER QUAD G550			
DR. CJH		DOAS REPLACEMENT			
CHK. CJM					
SUBMITTED BY: CJM		MECHANICAL ADMIN			
DESIGN DIR.C. Moos, PE		DOAS DDC POINTS			
APPROVED: PWO OR OICC		DATE	SIZE	CODE IDENT. NO	NAVFAC DRAWING NO.
			E1	80091	60042375
SATISFACTORY TO:		DATE	CONST. CONTR.		N40085-25-B-0080
			SCALE: NOTED		SPEC. 05-25-0080
					SHEET 8 OF 8